

Urban forms and flood resilience: Evaluating the impact of diverse micro-scale configurations on water flow dynamics

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RECEIVED 18.12.2025

ACCEPTED 22.06.2026

AVAILABLE ONLINE 17.09.2026

Highlights:

- Block-scale urban forms systemically influence flow characteristics.
- Urban form parameters correlate with inundation depth, velocity and hazard.
- Building coverage significantly impacts the flow characteristics.
- Practical insights are proposed for developing water-sensitive urban forms.

Abstract: Urban flooding has become an increasingly critical challenge worldwide, driven by urbanisation-induced changes to natural drainage systems and intensifying climate-driven precipitation extremes. Previous research on water-sensitive urban design has largely focused on stormwater runoff in simplified urban layouts, while giving less attention to the complexity of real urban form and its effect on flood behaviour. This study addresses this gap by systematically evaluating how diverse and complex urban configurations influence flood flow characteristics. Eight representative urban layouts, including four procedurally generated forms and four realistic forms, were developed within a two square kilometre area (1×2 km) using UrbanXTool, OpenStreetMap data, and ArcGIS. Two-dimensional hydraulic modelling was conducted under uniform hydraulic conditions to test the influence of six urban morphological parameters: street orientation, street curvature, street length, street width, mean parcel area, and building setbacks. A mild 1% slope, typical of floodplain environments, was selected to isolate the effect of urban form while limiting external hydrological variability. The results indicate that mean parcel setbacks (−0.64), mean parcel area (0.41), and street orientation (−0.21) are the parameters most strongly associated with changes in flood dynamics. High-density urban layouts with irregular street networks resulted in a 35.65% increase in peak water depth and

a 24.30% increase in the building damage rate compared with other configurations. These findings provide actionable insights for designing flood-resilient urban environments and advance current understanding of how urban morphology influences flood risk.

Keywords: 2D simulation, flood resilience, micro scale, spatial planning, urban form

INTRODUCTION

Climate change and rapid urbanisation are increasing urban flood risk, with important consequences for the built environment and socioeconomic systems (Mabrouk *et al.*, 2024; Mabrouk, 2026; Biswash *et al.*, 2026). Although flood management has traditionally relied on structural measures, increasing attention is being given to complementary planning-based and non-structural strategies. Löwe *et al.* (2017), using urban-development and hydrodynamic modelling in Melbourne, demonstrated that planning policies can contribute effectively to flood-risk reduction under alternative development and climate scenarios. Similarly, Mustafa *et al.* (2018) showed that different urban expansion and densification scenarios in Wallonia, Belgium, can substantially influence future flood exposure and damage. These findings emphasise the role of spatial planning and urban form in flood-resilient development.

This perspective is consistent with the broader development of integrated urban stormwater-management approaches, including Low Impact Development, Sustainable Urban Drainage Systems, Water-Sensitive Urban Design, and the Sponge City concept (Dong *et al.*, 2018; Qin, 2020; Luo *et al.*, 2021; Yin *et al.*, 2021). These approaches differ in scope and implementation but generally aim to improve urban stormwater retention, storage, infiltration, conveyance, and management. The present study focuses specifically on one planning dimension within this broader context: how the geometric organisation of buildings, parcels, and streets influences surface-flow behaviour under controlled hydraulic conditions.

Previous research demonstrates that urban morphology can modify floodwater storage, obstruction, conveyance, and flow direction. Experimental evidence has been obtained under several hydraulic conditions. Tomiczek *et al.* (2016), although investigating tsunami-induced rather than pluvial flooding, demonstrated that the arrangement and setback of buildings can influence local hydrodynamic conditions and structural loading. Huang *et al.* (2014) combined flume experiments and two-dimensional modelling to investigate building coverage and proposed an adjustment to Manning's roughness coefficient to represent building-induced blockage. Velickovic, Zech and Soares-Frazão (2017) examined the effects of street width and street orientation relative to the dominant flow direction using idealised laboratory configurations and developed an anisotropic porosity representation of urban flow. Dong *et al.* (2021) combined laboratory experiments and numerical modelling to investigate flash-flood inundation over an idealised urban street and showed that both surface configuration and the underground drainage system influence flood behaviour.

At the urban-block scale, Li *et al.* (2021b) combined computational modelling with laboratory validation to investigate long-duration flooding across different urban forms. Their results showed that conveyance porosity in the principal flow direction

strongly affects upstream water depth, downstream discharge distribution, and flow exchange through urban blocks. Rainfall-driven experiments provide complementary evidence. Cea, Garrido and Puertas (2010) conducted experiments using eight simplified urban configurations primarily to validate two-dimensional rainfall-runoff models, while Isidoro, Lima de and Leandro (2013) demonstrated that rooftop connectivity and spatial arrangement influence runoff hydrographs, with more connected configurations generally producing lower peak discharges and longer runoff durations.

Systematic computational studies have extended these investigations across larger sets of urban configurations. Bruwier *et al.* (2018) analysed 2,000 synthetic urban patterns and demonstrated that inundation depth is influenced by multiple morphological characteristics, including building coverage, parcel dimensions, and the organisation of inter-building spaces. Their results indicated that urban configurations with greater opportunities for flow conveyance between buildings can reduce upstream water-depth increases. Mustafa *et al.* (2020) subsequently integrated procedural urban-layout generation, hydraulic simulation, and optimisation to explore flood-sensitive urban configurations. Under pluvial-flooding conditions, Bruwier *et al.* (2020) analysed 2,000 synthetic urban forms and showed that building coverage strongly influences flood behaviour, while building arrangement affects water depth more substantially than stored water volume. More recently, Balaian, Sanders and Abdolhosseini Qomi (2024) demonstrated that urban flood response depends on both urban porosity and the spatial organisation of buildings, reinforcing the importance of considering configuration rather than density alone.

Studies conducted at broader urban scales further support the relationship between spatial structure and flood response. Lin *et al.* (2021) used a random-forest algorithm to examine three-dimensional building configuration and identified building density, building congestion degree, and building coverage ratio as important predictors of pluvial flooding. Lin, Shaad and Girot (2016) integrated landscape and hydrodynamic modelling to assess alternative rehabilitation scenarios for the Ciliwung River in Jakarta. Mohammed (2021) investigated non-structural flood-risk mitigation in Addis Ababa and found that aligning street networks with the dominant stormwater-flow direction had relatively little effect on total runoff volume but could improve stormwater storage and subsequent drainage compared with more perpendicular network arrangements.

Collectively, these studies demonstrate that urban flood dynamics are influenced by interconnected morphological characteristics, including building coverage, parcel dimensions, building setbacks, directional conveyance porosity, street width, street orientation, and the spatial arrangement of buildings and open spaces. However, important methodological differences remain. Laboratory investigations generally use simplified geometries, whereas large parametric studies predominantly rely

on synthetic or procedurally generated configurations. Conversely, studies based on real urban environments incorporate greater morphological complexity but make it more difficult to isolate the influence of urban form from differences in topography, rainfall, drainage infrastructure, land cover, and other site-specific conditions.

Therefore, the remaining research need is not simply to establish whether urban form affects flooding, as this relationship is already well recognised, but to examine whether relationships identified from controlled and synthetic configurations remain evident in morphologically more complex urban forms under comparable hydraulic conditions. A controlled comparison between realistic and procedurally generated layouts can help isolate the hydraulic influence of urban morphology while limiting interference from site-specific factors.

Accordingly, this study evaluates eight contrasting urban configurations, comprising realistic and procedurally generated forms, under identical two-dimensional hydraulic conditions. Six planning-relevant morphological characteristics are considered: mean parcel area, building setbacks, street width, street curvature, street length, and street orientation. The study tests the hypothesis that variations in these characteristics are systematically associated with differences in water depth, flow velocity, discharge, and hydraulic hazard. Specifically, the objectives are to quantify hydraulic differences among the eight urban forms, evaluate the relative influence of the six morphological parameters, assess the stability of these relationships under alternative hydraulic conditions, and derive planning-relevant insights for the development of more flood-resilient urban configurations.

MATERIALS AND METHODS

METHODOLOGICAL FRAMEWORK

The methodology of this study encompasses three primary steps (Fig. 1). Initially, procedural modelling of urban forms is conducted, focusing on eight configurations generated using six parameters, categorised into realistic and procedurally generated forms. In this study, the term “procedurally generated” refers to urban layouts automatically created using rule-based modelling techniques to explore plausible yet non-existing configurations, in contrast to “realistic” layouts derived from actual urban areas. This process utilises UrbanXTool and OpenStreetMap (OSM) in conjunction with the Geographic Information System (GIS). Subsequently, two-dimensional numerical simulations are performed for each urban form under identical hydraulic boundary conditions through steady flow, employing IBER 3 software. This software offers high modelling flexibility and the capability to assess multiple design scenarios using its built-in optimisation algorithms. The final step involves a statistical analysis of the influence of urban form parameters on flow characteristics. This comprehensive approach culminates in presenting the findings, detailing the generated building layouts and the impact of urban characteristics on flood dynamics and an examination of their sensitivity. This thorough analysis enhances understanding of urban flooding mechanisms and contributes valuable insights for future urban planning and flood management strategies.

AUTOMATED GENERATION OF URBAN MORPHOLOGIES

Blocks are used for larger-scale urban planning assessments, as they represent groups of buildings and provide insights into overall urban structure and connectivity. Parcels, smaller units,

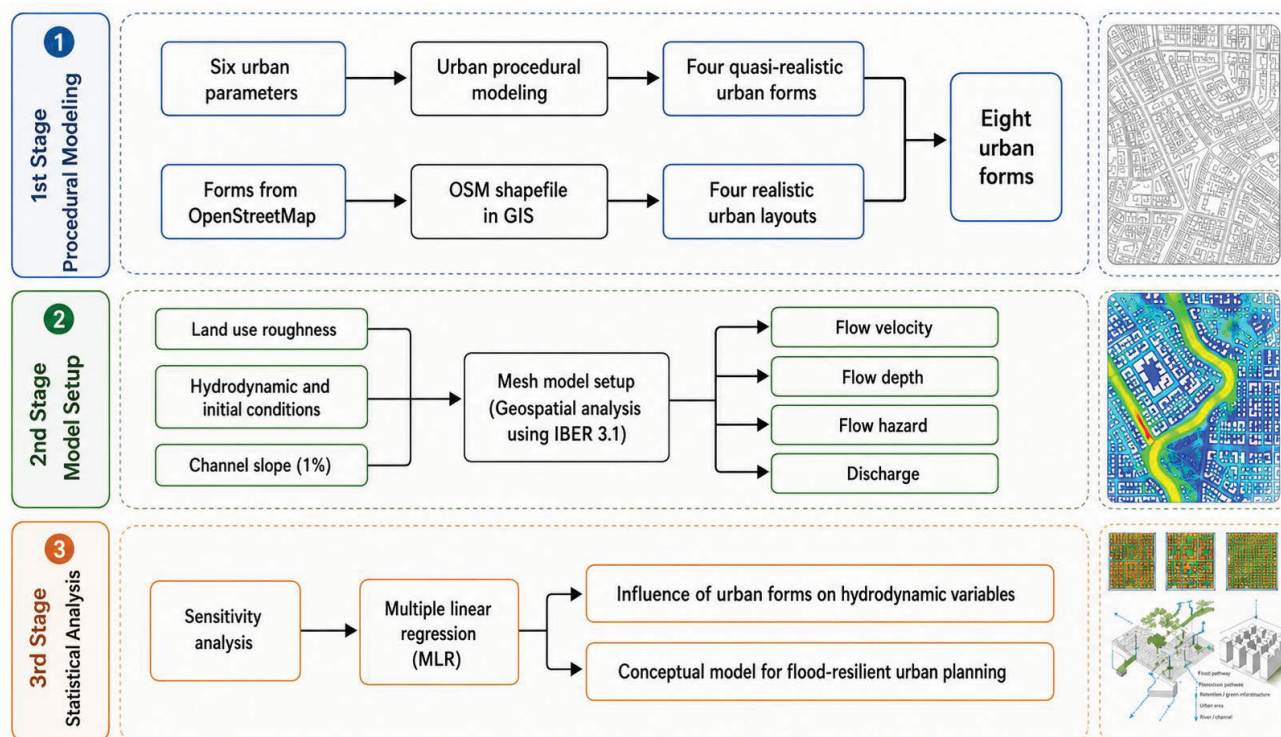


Fig. 1. Three-step methodological framework used to generate urban layouts, perform hydraulic simulations, and assess the sensitivity of flood responses to urban form parameters; OSM = OpenStreetMap, GIS = Geographic Information System; source: own elaboration

are employed for more detailed, property-level analyses that consider individual land use and building configurations. Merging between blocks and parcels allows the capture of different aspects of urban form that influence flood resilience, such as street network patterns, building density, and open space distribution. Furthermore, this comprehensive approach ensures that our study represents a realistic spectrum of urban environments, providing a robust foundation for assessing flood resilience across various urban typologies. Block-scale procedural modelling of urban forms involves the automated generation of detailed blocks using specific parameters and rules. This process includes creating a street network skeleton based on street length, orientation, and curvature. Parcels within the block are then delineated considering street widths and mean parcel area, as shown in Figure 2. Finally, building footprints and setbacks precisely place structures within each parcel, efficiently creating diverse and plausible urban layouts that can be easily modified and optimised. In this study, an enhanced version of the methodology proposed by Bruwier *et al.* was employed to accommodate a greater diversity in street network configurations (Mustafa, 2018; Bruwier *et al.*, 2020). UrbanXTool has been used as an open-source Grasshopper plugin for Rhino, including tools for network structure generation, regulatory planning, urban design, auto-generation, and various urban analyses.

Based on the previous urban parameters, this study examines four realistic urban forms and four procedurally generated ones. The realistic forms serve as benchmarks, representing actual urban environments and capturing the complexity and diversity of real-world urban systems. The selection of realistic urban forms for assessing flood simulations considered several key criteria to ensure representativeness and accuracy. The selection considered areas with known flood histories or vulnerabilities to validate and improve the simulation models. These forms were carefully selected from various cities using OSM, a comprehensive open-source geospatial dataset. The OSM files of the chosen layouts were then exported to GIS software to obtain a range of urban layouts with varying characteristics. The procedurally generated urban forms were generated using advanced urban modelling techniques, explicitly employing UrbanXTools to explore hypothetical yet plausible urban configurations. These forms were automatically generated at 2 km², allowing for the exploration of novel or idealised urban layouts that may not exist and represent innovative solutions to urban challenges.

The generated urban forms, which differed in variable values, are shown in Figure 3. Layouts 6 and 7 are high-density forms, whereas layouts 1, 2, 3, 4, 5, and 8 are low-density, varying from planned to random. Regarding the selection of random

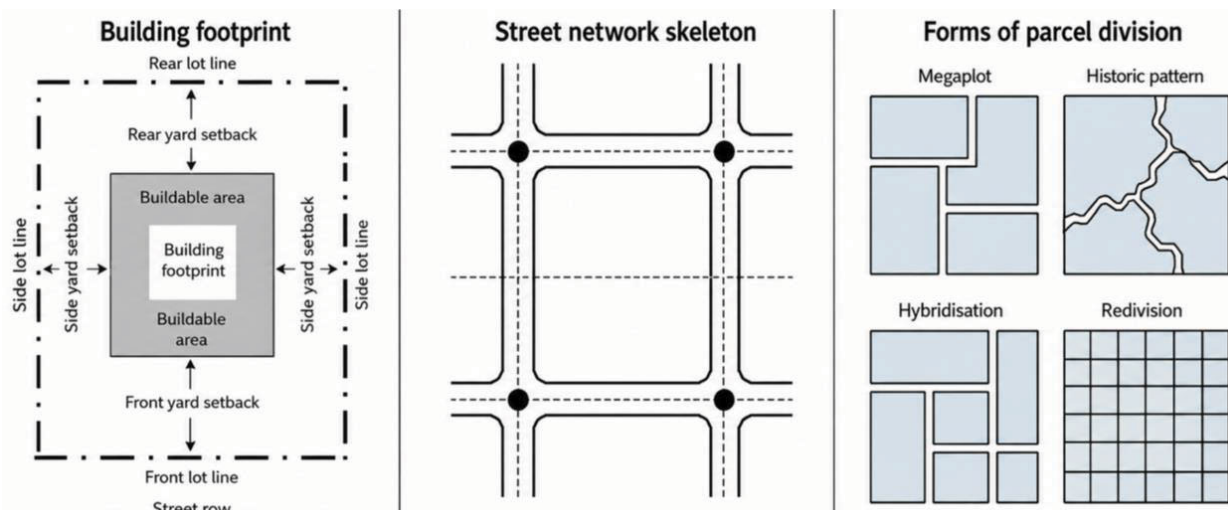


Fig. 2. Main components of the automated urban form generation process, including street skeletons, parcel division, and building footprints; ROW = right-of-way; source: own elaboration

Six input parameters of practical significance in urban planning are incorporated, encompassing the characteristics of streets and buildings (Tab. 1). For example, the mean parcel/block area, which ranges from 180 to 7000 m², dictates the size and density of individual lots within a block, impacting the overall building footprint. Parcel setbacks, ranging from 0 to 6 m, influence the distance between buildings, streetscape, and potential for green buffers (Mabrouk *et al.*, 2023; Mabrouk *et al.*, 2024). The minimum value for the side setback is set at 0 metres, as many buildings globally do not incorporate any setbacks. Previous studies have not considered this factor during simulations (Bruwier *et al.*, 2018). The street width, ranging from 4 to 40 m, determines the scale of roadways and influences the water flow. Street curvature, with values between 0 and 15 km⁻¹, and street length, ranging from 150 m to 1800 m, shape the geometric configuration of the street network.

Table 1. Six urban form parameters (Pa. 1–Pa. 6) used to generate the procedurally generated layouts and assess their influence on flood flow dynamics

Parameter code	Parameter	Range
Pa. 1	mean parcel/block area	180–7000 m ²
Pa. 2	parcel setbacks	0–6 m
Pa. 3	street width	4–40 m
Pa. 4	street curvature	0–15 km ⁻¹
Pa. 5	street length	150–1800 m
Pa. 6	street orientation	0°–180°

Source: own elaboration based on Bruwier *et al.* (2018), Mustafa *et al.* (2019), Li *et al.* (2021b).

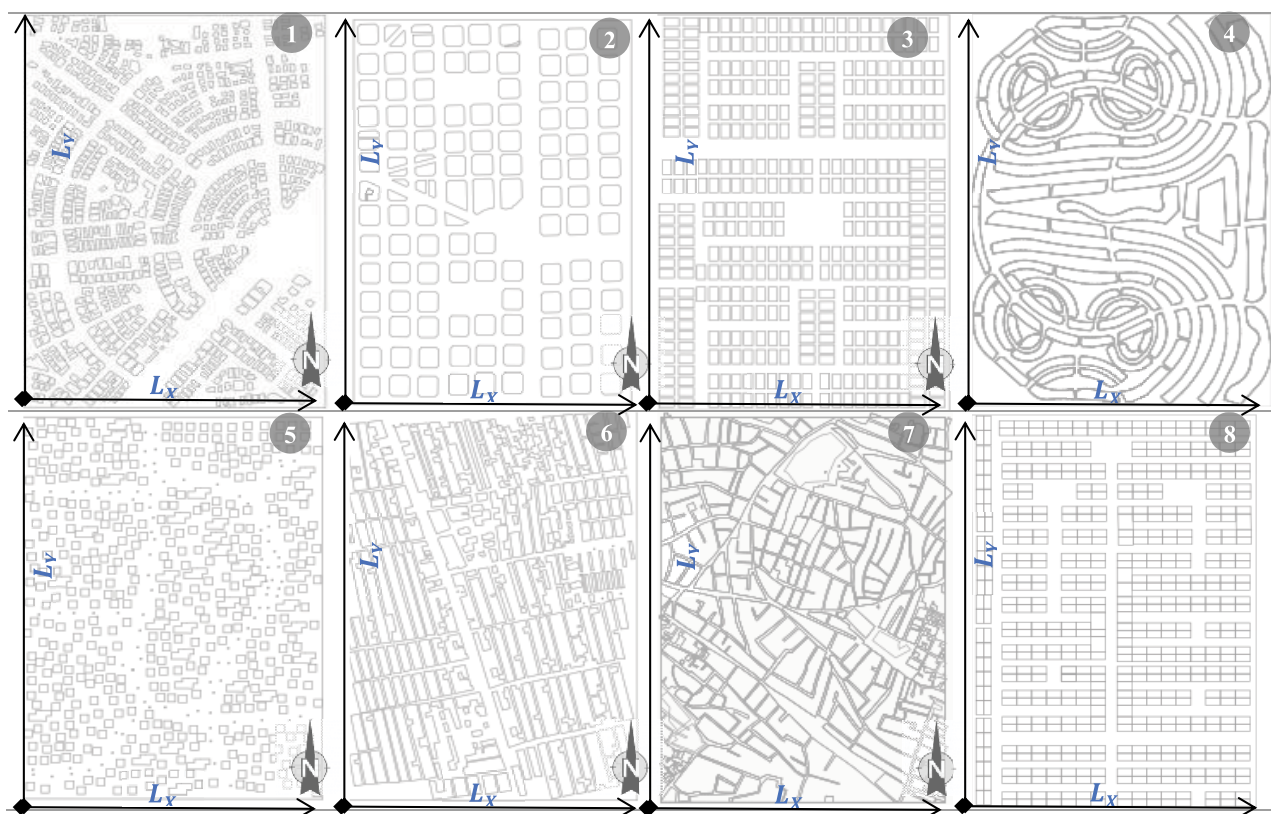


Fig. 3. Eight urban forms of realistic and procedurally generated layouts: 1) realistic form, 2) procedurally generated form, 3) realistic form, 4) procedurally generated form, 5) procedurally generated form, 6) realistic form, 7) realistic form, 8) procedurally generated form; L_x and L_y = a region's length in the x and y directions, respectively; source: own elaboration

urban forms, most developing countries exhibit this form in their cities, providing policymakers and urban planners with a proactive vision of potential risk mitigation. Layouts differ in their variable values; for example, the average street length in layout 4 is about 1.77 times that of layout 2. The average street widths in layouts 1, 2, and 5 are almost three times higher than in layouts 6 and 7. Layouts 4, 6, and 7 show substantially larger building blocks than layouts 1, 2, and 3. In layouts 2, 3, 6, and 8, the building alignment directs the flow from the west upstream face to the north downstream face. In contrast, layouts 1, 4, and 7 promote flow connections from the west upstream face to the

east downstream face. Different models with varying setbacks are proposed, such as layouts 1, 3, and 5, which have setbacks of 2 m, 3 m, and 4 m, respectively, while layouts 6 and 7 have no setbacks.

A comprehensive overview of urban layouts, highlighting key metrics that characterise each layout's design and spatial configuration, is presented in Table 2. For instance, layout 1 features a mean parcel area of 180 m² with a mean setback of 2 m. The street width is 18 m, while the street length is 1,200 m, and a building coverage ratio is 61%. In contrast, layout 7 stands out with the largest mean parcel area of 7000 m² and has no setbacks and minimal street widths (4–6 m for minor streets).

Table 2. Morphological characteristics of the eight simulated urban layouts (La.), highlighting differences in parcel size, street dimensions, setbacks, and building coverage

Layout	Mean parcel area (m²)	Mean parcel setbacks	Major street width	Minor street width	Street length	Building coverage ratio (%)
		m				
La. 1	180	2	18	10	1200	61%
La. 2	5000	3	25	13	970	65%
La. 3	410	3	30	15	970	68%
La. 4	6000	6	19	12	1720	54%
La. 5	250	4	33	15	970	48%
La. 6	6500	0	13	6	900	82%
La. 7	7000	0	6	4	1120	90%
La. 8	270	0	8	6	1354	76%

Source: own elaboration.

Layouts 5 and 6 demonstrate varying configurations, with layout 5 having a mean parcel area of 250 m² and a building coverage ratio of 48%. In comparison, layout 6 features no setbacks and a street width of 13 m.

HYDRAULIC ANALYSIS OF DIVERSE URBAN MORPHOLOGIES

A simplified hydraulic approach for modelling urban flood dynamics

The second phase employs a computationally efficient hydraulic simulation model to analyse the flow dynamics across all urban forms under consistent boundary conditions, ensuring comparability and reliability of the results. The urban built environment profoundly influences flood dynamics through three primary mechanisms: (1) the reduction of water storage capacity due to building presence, which is modelled using an isotropic storage porosity parameter ϕ in continuity and momentum equations, (2) the channelisation of water flow by urban structures, resulting in complex hydrodynamic features that are modelled by adjusting friction parameters and introducing drag coefficients, and (3) the redirection of floodwaters by buildings acting as obstacles (Guinot and Soares-Frazão, 2006; Guinot, Sanders and Schubert, 2017; Velickovic, Zech and Soares-Frazão, 2017). These effects fundamentally alter water behaviour in urban landscapes, significantly impacting water accumulation, movement, and flow directions compared to unobstructed terrain. In response to these challenges, two-dimensional shallow-water hydraulic models have emerged as the state-of-the-art approach for simulating urban inundation flow (El Kadi Abderrezzak, Paquier and Mignot, 2009; Ghostine *et al.*, 2015). When accounting for obstacles in inundation modelling, three main approaches have been widely identified in the literature: increasing the roughness parameter, representing obstacles as mesh voids, or employing a porosity-based model (Schubert and Sanders, 2012; Kim and Choi, 2015; Özgen *et al.*, 2016; Bruwier *et al.*, 2017).

Two compatible approaches are considered: increasing the roughness parameter and representing obstacles as mesh voids. These methods maintain model simplicity while effectively capturing the impact of urban features on flood flow. In the present implementation, storage and conveyance porosity were not explicitly parameterised, and all ground surfaces were assumed to be impervious (asphalt). This simplification was adopted to isolate the influence of urban morphological characteristics under controlled hydraulic conditions. This assumption does not account for surface permeability or blue-green infrastructure (BGI); therefore, the results should be interpreted as a comparative assessment rather than a direct representation of real-world water-sensitive urban systems. Nevertheless, the findings provide fundamental insights into the role of urban form in shaping flood dynamics and can inform future studies that integrate permeability and nature-based solutions.

The momentum equations in this model incorporate a drag loss term consistent with the formulation proposed by Schubert and Sanders (2012), accounting for the resistance flow encounters due to urban structures and obstacles. The bottom friction drag coefficient is parameterised using a Manning coefficient (n_m), so $c = \frac{g n_m^2}{h^{\frac{2}{3}}}$ (where: c = drag coefficient, g = gravitational acceleration, h = water depth). Manning coefficient $n_m = 0.04 \text{ s} \cdot \text{m}^{-\frac{1}{3}}$

value has been carefully selected to align with previous studies, particularly those conducted by Bazin (2013) and Bruwier *et al.* (2017). The chosen roughness coefficient is designed to encapsulate various sources of flow resistance typically encountered in urban environments, including traditional bottom friction and the cumulative effect of small-scale obstacles such as debris, parked vehicles, and street furniture. The building drag coefficient is computed like vegetative drag laws: $c_D = \frac{1}{2} c_D^0 a h$, (where: a = frontal area). The units of a are length (L^{-1}) corresponding to the frontal width of obstructions in Ω normalised by Ω . Dimensionless obstruction drag coefficient, represented by c_D^0 , values for 2D flows and shaped obstructions are tabulated in textbooks as c_D^0 is approximately from 2 to 3.

Iber 3.1 model set-up

The Iber model setup integrates geometry and hydrodynamics. Although land-use categories are geometrically represented, no land-use-based roughness differentiation was applied. The model's geometry was established by defining the spatial layout of the area under study, which includes detailed representations of streets, open spaces, and buildings. This involves creating a mesh that discretises the computational domain into smaller elements, allowing for precise calculations of flow dynamics. The model utilised unstructured meshes composed of triangular elements, depending on the complexity of the urban forms. This flexibility ensures that the model can accurately capture the effects of urban form on flood characteristics.

Furthermore, to ensure strict comparability between layouts and to isolate the influence of urban morphology, all ground surfaces were assigned a uniform Manning roughness coefficient corresponding to asphalt conditions. Buildings were represented as impermeable obstacles through mesh void treatment and associated drag coefficients. No land-use variability was introduced across layouts, thereby ensuring that differences in flood behaviour arise exclusively from geometric and morphological characteristics rather than material heterogeneity. In addition to geometry and land use, hydrodynamic conditions were meticulously defined to ensure realistic simulations. The Iber model employs initial and boundary conditions tailored to the analysed urban flooding scenario. This includes setting up inlet and outlet conditions that reflect real-world hydrodynamics, such as supercritical and subcritical flow regimes. The calculation data settings specify key simulation parameters, including time steps and output frequency, which are crucial for capturing transient flood behaviours accurately. Once the simulation is executed, the Iber model generates results detailing flood characteristics such as water depth, flow velocity, and discharge rates.

Discrete form and spatial discretisation

To streamline calculations, Guinot, Sanders and Schubert (2017), Bruwier *et al.* (2017), and Ferrari *et al.* (2019) introduced different equations that are particularly useful for simulating water flow in urban environments. Over a computational cell (j), $\langle uh \rangle_j$ and $\langle vh \rangle_j$ represent the cell-averaged unit discharges in the x - and y -directions, respectively, calculated in the same manner (Eq. 1) but using the corresponding velocity component.

$$\langle h \rangle_j = \frac{\int_{\Omega_j} i h d\Omega}{\int_{\Omega_j} i d\Omega} \quad (1a)$$

$$\langle uh \rangle_j = \frac{\int_{\Omega_j} i u h d\Omega}{\int_{\Omega_j} i d\Omega} \quad (1b)$$

$$\langle vh \rangle_j = \frac{\int_{\Omega_j} i v h d\Omega}{\int_{\Omega_j} i d\Omega} \quad (1c)$$

$$\langle u \rangle_j = \frac{\langle uh \rangle_j}{\langle h \rangle_j} \quad (1d)$$

$$\langle v \rangle_j = \frac{\langle vh \rangle_j}{\langle h \rangle_j} \quad (1e)$$

$$\langle V \rangle_j = \sqrt{\langle u \rangle_j^2 + \langle v \rangle_j^2} \quad (1f)$$

where: Ω = the computational cell; i is an indicator function (1 in the wet area, 0 elsewhere), h = the local water depth, d = the differential area element for integration, u = the local x -direction velocity, v = the local y -direction velocity, V = the resultant velocity magnitude.

Based on the characteristics of the flux vector splitting method, the model implements a downstream discretisation of the bottom slope and a mean evaluation of the corresponding water heights, as shown in Equation (2) (Erpicum *et al.*, 2010).

$$-gh \frac{\partial z_b}{\partial x, y} \Big|_i \rightarrow -g \frac{(h_{|i+1} - h_{|i})}{2} \frac{\partial (z_{b|_{i+1}} - z_{b|_i})}{\partial x, y} \quad (2)$$

where: g = gravitational acceleration, z_b = the bottom elevation, $i + 1$ = downstream mesh along x or y axis.

Triangular elements instead of grid meshes were used, allowing for flexible adaptation to irregular urban geometries, as shown in Figure 4. This is particularly advantageous when modelling the four realistic urban forms presented in the study. The quality criterion measures the likeness of the element to the reference one (an equilateral triangle in the case of triangles). Its value is 1 for a perfect element (the reference one), decreasing as it worsens. If it reaches a negative figure, it means that the element has a negative Jacobian at some point. The shape quality (q) of triangles is measured as follows (GiD_17, 2024):

$$q = \frac{4 \cdot \sqrt{3} \cdot A}{\sum_{i=1}^3 l_i^2} \quad (3)$$

where: A = area of the triangle, l_i ($i = 1, \dots, 3$) = lengths of the triangle's edges.

A variety of triangular mesh configurations were evaluated, beginning with a uniform element size of 6 m and progressing to a finer mesh with an element size of 4 m. This approach aimed to accurately discretise the topography of the study area, ensuring precise representation of the terrain and enhancing the model's ability to simulate hydrodynamic behaviour effectively. Mesh elements vary across layouts based on urban morphological parameters, with a minimum of 15,059 cells for each configuration.

Boundary conditions of urban forms

Model boundary conditions for assessing urban forms and flows are crucial for accurately simulating urban flood dynamics. In this study, the upstream boundary (2D inlet) is located on the southern side of the computational domain. In contrast, the

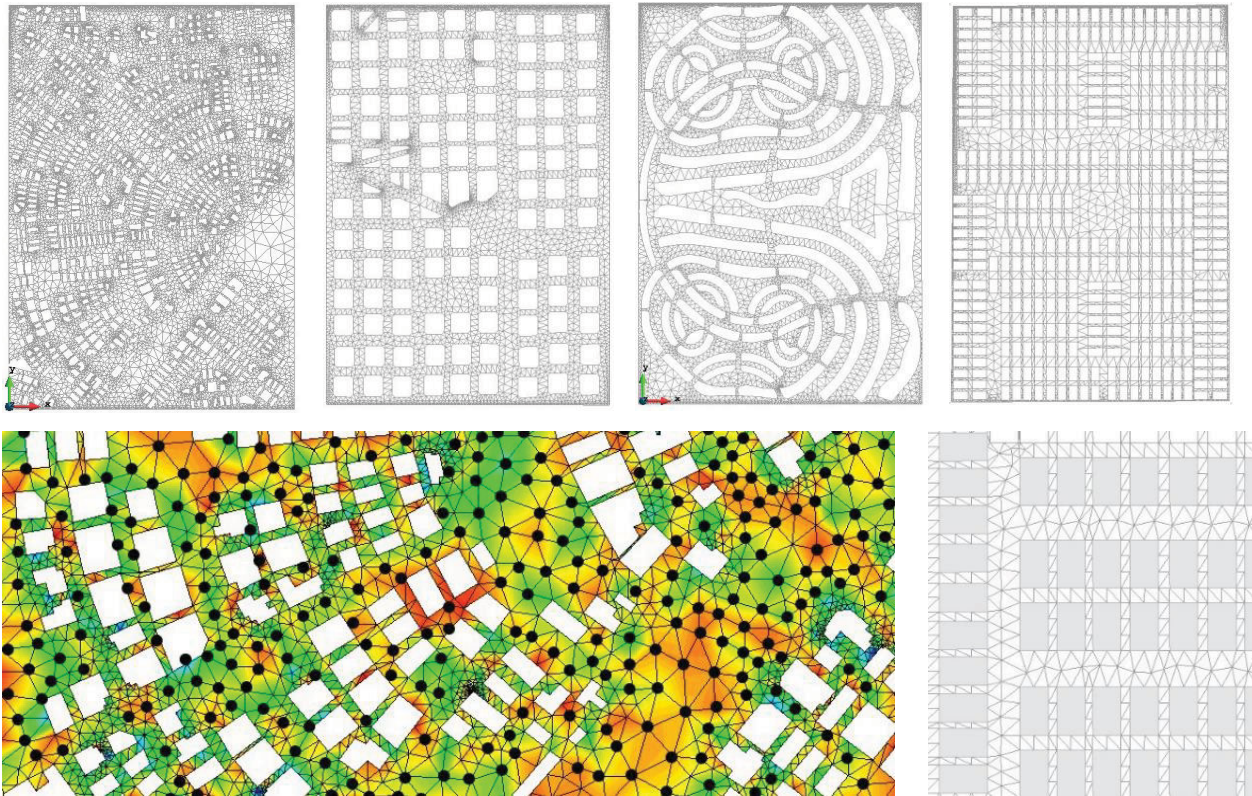


Fig. 4. Sample triangular meshes used in the numerical simulations, illustrating adaptation of the computational grid to different urban geometries; source: own elaboration

north, east, and west sides are the downstream boundaries (2D outlet), as shown in Figure 5. In the model setup, a 25-metre-wide strip devoid of buildings was maintained along the upstream boundaries of the computational domain to ensure proper inflow distribution, allow for a smooth transition from the unobstructed upstream flow to the complex urban flow patterns, and prevent immediate flow obstruction at the inlet. A total steady inflow discharge (SID) of $300 \text{ m}^3 \cdot \text{s}^{-1}$ along the upstream sides has been prescribed. On the downstream boundaries of the model, the outflow discharge (q_j) for each cell (j) was determined as follows in Equation 4 (Bruwier *et al.*, 2018):

$$q_j = C_1 \Delta x \sqrt{2g(\langle h \rangle_j - C_2)^3} \quad (4)$$

where: Δx = the triangular spacing (the distance between grid points in a triangular mesh used for spatial discretisation of the model domain), C_1 and C_2 = constants respectively equal to 0.5 and 0.3.

force (D), which represents the equivalent water depth at rest that would produce a static force equal to the total force exerted by the flowing water. This method, developed and refined by Aureli *et al.* (2008) and further enhanced by Cea Gómez *et al.* (2019) and Ferrari *et al.* (2019), Ferrari and Viero (2020), is expressed as follows:

$$D = h \sqrt{1 + 2\text{Fr}^2} \quad (6)$$

According to the partition shown in Figure 6, four hazard ranks are defined as follows (Ferrari *et al.*, 2019; Ferrari and Viero, 2020): low ($0 \leq D < 0.5$), medium ($0.5 \leq D < 1$), high ($1 \leq D < 1.5$) and very high ($D \geq 1.5$).

Statistical interpretation of sensitivity analysis

The preceding analysis yielded a comprehensive dataset detailing flow characteristics across the computational mesh, encompassing water depth and flow velocities within more than 15,000 cells. This extensive dataset was synthesised into a single indicator,

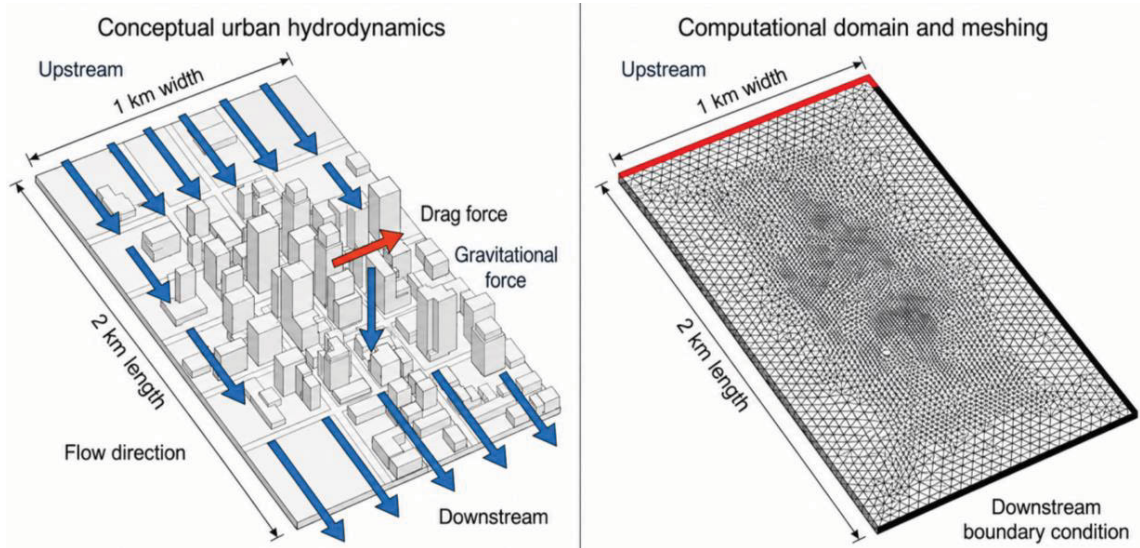


Fig. 5. Hydraulic boundary conditions applied to all layouts, including the southern upstream inflow and the northern, eastern, and western downstream outlets; source: own elaboration

Froude number (Fr) is a dimensionless parameter that characterises the fluid flow ratio of the inertial to gravitational forces. The number (Fr) determines the flow regime using Equation 5: when $\text{Fr} < 1$, the flow is considered subcritical, indicating that waves can travel upstream; when $\text{Fr} > 1$, the flow is supercritical, meaning it is fast and waves cannot propagate upstream; and when $\text{Fr} \approx 1$, it represents critical flow conditions where the velocity of the flow matches the wave speed. The downstream boundary conditions result in Fr , ranging from 0.1 to 0.70 within the simulation domain.

$$\text{Fr} = \frac{V}{\sqrt{gh}} \quad (5)$$

To assess hydraulic hazard and severe inflow conditions in urban flooding scenarios, an approach employing Equation 6 is utilised. This equation incorporates a hydraulic hazard indicator for hydrostatic and dynamic loads, comprehensively evaluating flood dynamics. The indicator is quantified as the maximum total

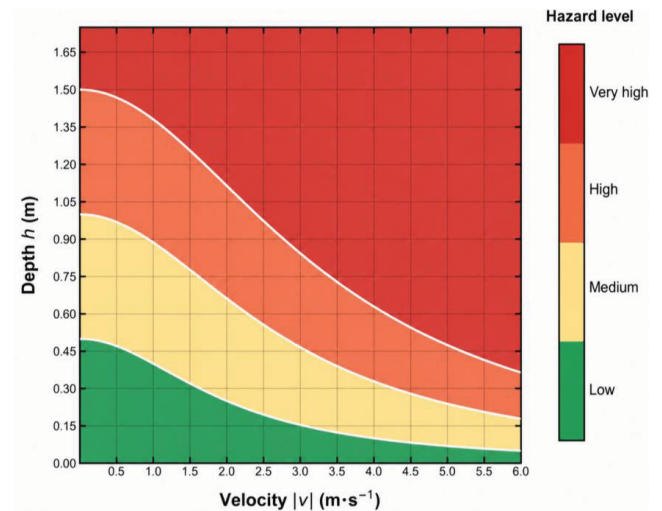


Fig. 6. Hazard classification based on maximum total force, distinguishing low, medium, high, and very high flood hazard levels; source: own elaboration

denoted as y , which quantifies flood severity for each layout examined to facilitate subsequent analyses.

The focus was placed on the increase in the 90th percentile of computed water depths along the upstream boundary of the domain, referred to as Δh_{90} , compared to a baseline scenario without any buildings. This specific metric indicates the overall flow resistance or loss of flow conveyance attributable to the presence of buildings within the area.

The 90th percentile water depth was selected instead of maximum depth to avoid sensitivity to localised numerical peaks or isolated cells that may not represent the overall hydraulic response of the layout. Compared to spatially averaged depth, the 90th percentile better captures high-severity flow conditions while remaining statistically robust. Flooded area and hazard exceedance indicators were not adopted as primary variables because the objective of the study is to isolate the hydraulic resistance effect induced by urban morphology rather than to quantify absolute damage or exposure. Focusing on the upstream boundary allows for a consistent comparison of backwater effects generated by different urban configurations, as this location directly reflects the cumulative obstruction and conveyance resistance imposed by the layout before flow redistribution occurs downstream. To facilitate subsequent analysis, the results are statistically analysed using standardised variables (z -scores), as shown in Equation 7:

$$\bar{x}_i = \frac{x_i - x_{i(\text{mean})}}{x_{i(\text{std})}}, \text{ and } \bar{y}_i = \frac{y_i - y_{i(\text{mean})}}{y_{i(\text{std})}} \quad (7a)$$

$$\mathbf{X} = \begin{bmatrix} \bar{x}_1^1 & \bar{x}_1^2 & \dots & \bar{x}_1^N \\ \bar{x}_2^1 & \bar{x}_2^2 & \dots & \bar{x}_2^N \\ \vdots & \vdots & \ddots & \vdots \\ \bar{x}_N^1 & \bar{x}_N^2 & \dots & \bar{x}_N^N \end{bmatrix} \text{ and } \mathbf{Y} = \begin{bmatrix} \bar{y}^1 \\ \bar{y}^2 \\ \vdots \\ \bar{y}^N \end{bmatrix} \quad (7b)$$

where: $\bar{x}_i$ represents the input variables ($i = 1$ to 6) describing various aspects of urban form or building layout characteristics, $\bar{y}^N$ = output variable (water flow characteristics), N = the total number of observations, $x_{i(\text{mean})}$, $x_{i(\text{std})}$, $y_{i(\text{mean})}$, $y_{i(\text{std})}$ = mean and standard deviation of each input variable x_i , the matrix notations $\mathbf{X}$ and $\mathbf{Y}$ represent the values of the variables x_i and y_n (Bruwier et al., 2018). Using a multiple linear regression (MLR), the influence of each of the six variables x_i on the flood indicators y was determined. The regression model includes six standardised urban form parameters (Pa.1–Pa.6); therefore, all matrix

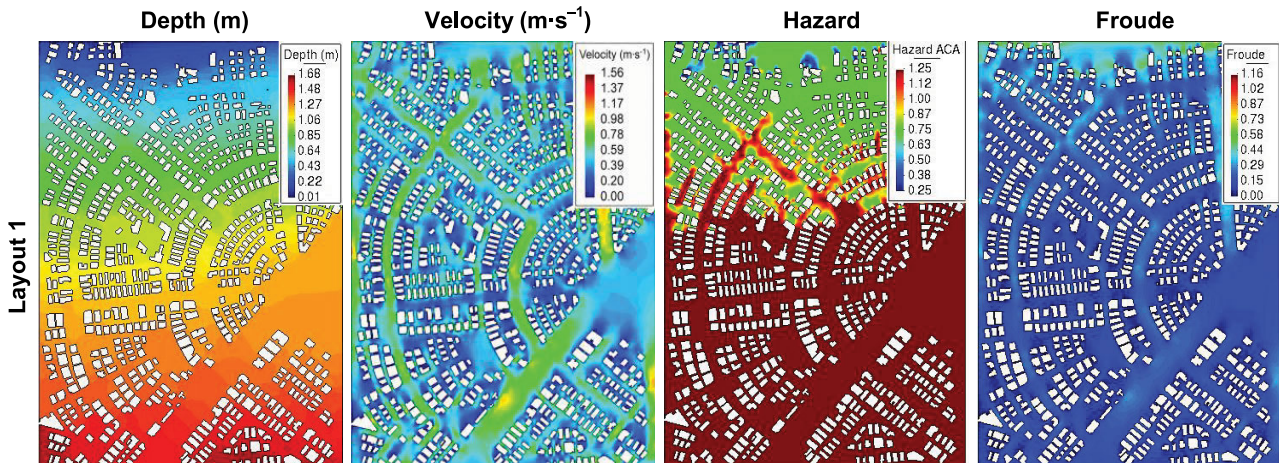
formulations and coefficient vectors are defined with six predictors. The regression analysis results yield the least squares linear coefficients, represented by the vector $\mathbf{A} = [a_1, a_2, \dots, a_6]^T$. These coefficients are computed as $\mathbf{A} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$. This equation illustrates how the coefficients quantify the sensitivity of the dependent variable y concerning each independent variable x_i . In addition to regression coefficients, Pearson correlation coefficients ρ_i are employed to assess the strength and direction of the linear relationship between each independent variable x_i and the dependent variable y . These coefficients are defined as $\rho_i = \frac{\text{cov}(x_i, y)}{x_{i(\text{std})} y_{\text{std}}}$ (where: $\rho_i = \text{cov}(x_i, y)$ = covariance between x_i and y ; $x_{i(\text{std})}$ and y_{std} = standard deviations of x_i and y , respectively).

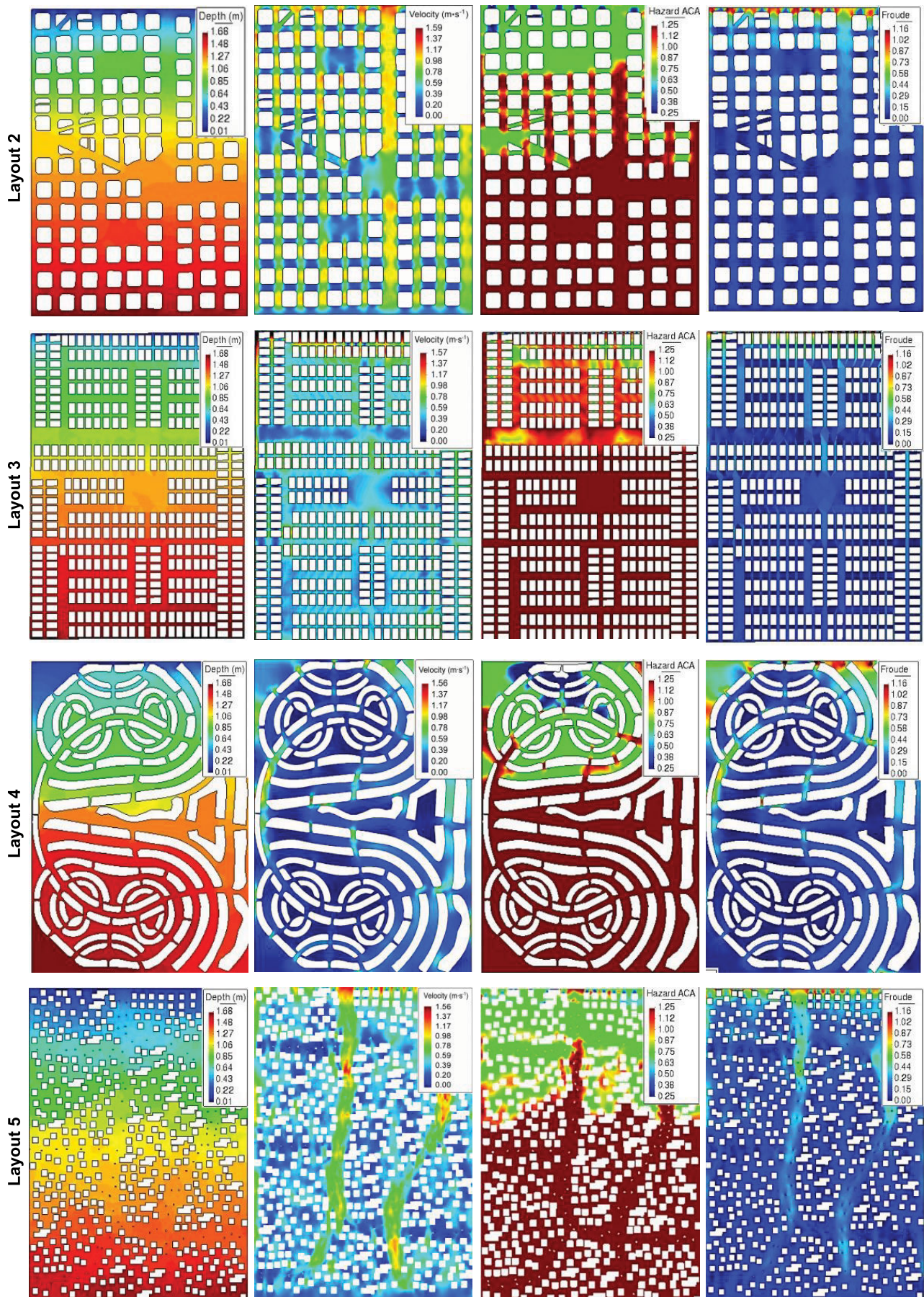
RESULTS

COMPUTED WATER INUNDATION CHARACTERISTICS

The simulations of 2D inundation were conducted to evaluate the effects of eight urban configurations on water depth, flow velocity, and flood hazard, as illustrated in Figures 7 and 8. The complete time-series results are provided in the Supplementary Materials (Tabs. S1 and S2). The software incorporated all numerical and descriptive data, including land use roughness and a simulation time of 4000 s for each urban form, deemed appropriate for flow progression from inlet to outlet. The analysis of flow depth indicated that maximum values are observed at the upstream locations, gradually decreasing as one moves downstream. Generally, the urban environment's flow is characterised by relatively low velocities, with a Froude number (Fr) less than 1. The maximum observed Fr does not exceed approximately 0.71, indicating that the flow remains subcritical.

The chosen flood level indicator (Δh_{90}) varies significantly along the upstream boundaries, ranging from 1.21 to 2.19 m. As different building layouts were analysed, it became apparent that changes in the arrangement and density of structures could substantially alter upstream flood levels. In addition to 90th percentile of computed water depths along the upstream boundary of the domain (Δh_{90}), a building damage rate indicator was computed. For each layout, building cells were identified from the mesh representation. A building cell was classified as damaged when the simulated water depth at that cell exceeded the corresponding depth obtained in the baseline configuration without buildings under identical hydraulic conditions. The building





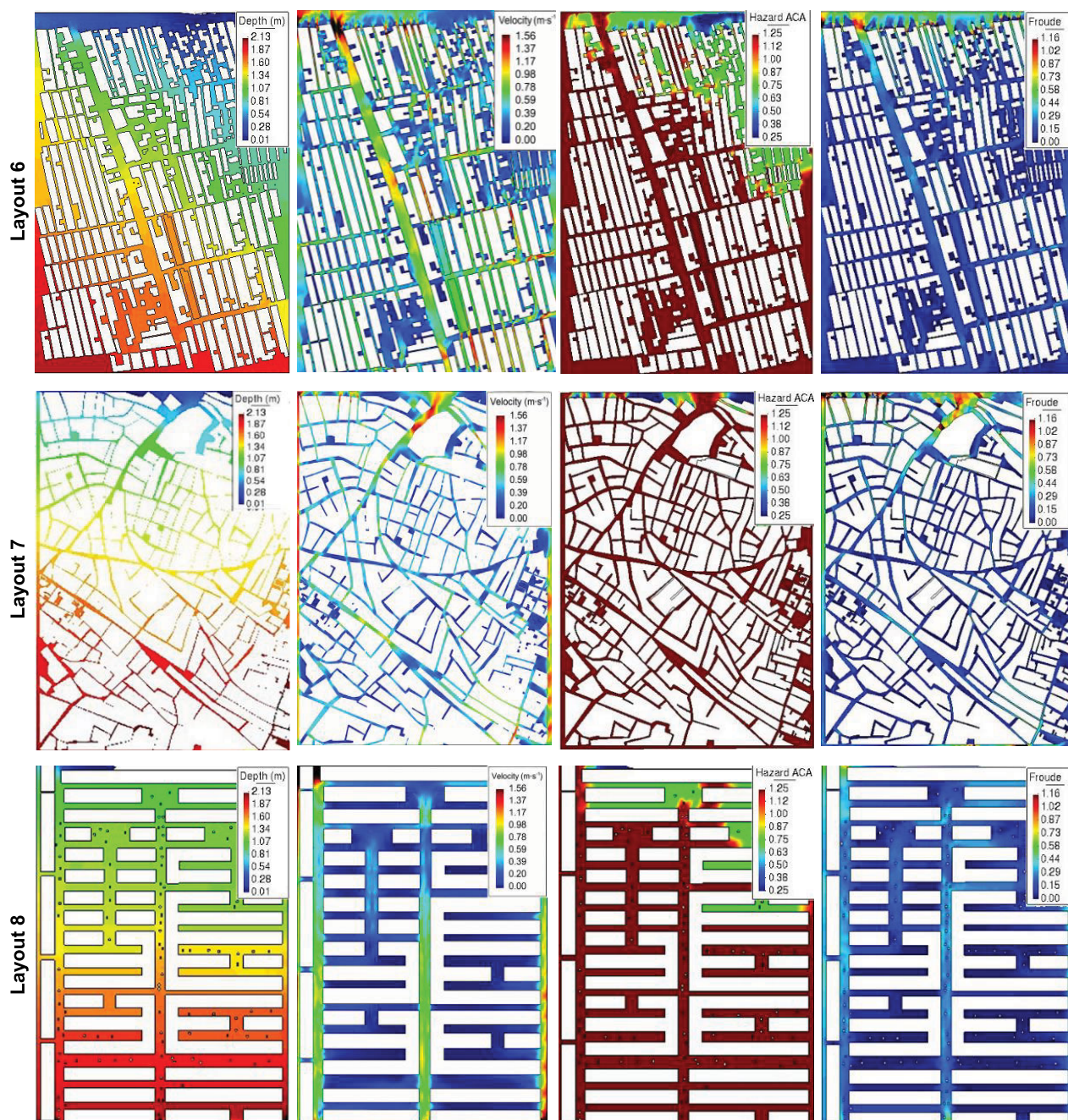


Fig. 7. Time evolution of flood depth, velocity, hazard, and Froude number across the eight urban layouts, showing marked hydraulic contrasts between configurations; source: own study

damage rate was then calculated as the ratio between the number of damaged building cells and the total number of building cells in the layout. This variation is particularly noticeable in layouts 1, 3, 4, and 5, which feature setbacks facilitating more intense flow exchanges between streets and void areas within building blocks. In layout 7, flow velocity and water depth experience notable increases, reaching values of $1.17 \text{ m}\cdot\text{s}^{-1}$ for velocity and 2.19 m for water depth. This pattern differs from the other layouts, where flow speed increased while water depth decreased during the simulation period. This behaviour is likely associated with greater water accumulation along the main streets, which may result from the narrow road network and the absence of setbacks or designated storage areas. Furthermore, the percentage of damaged buildings in this layout increased by 24% compared to other

configurations. Interestingly, despite the high building coverage in layout 2, the relatively vertical orientation of the downstream network facilitated a smoother outflow with fewer associated water hazards. In contrast, flow velocity significantly increased from the estuary before gradually decreasing over time for layouts characterised by vertical networks and lateral setbacks – such as layouts 1, 3, 4, and 5. The percentage of affected buildings decreased near the estuary because of lower water depths. Velocity remained relatively stable after approximately 1900 s in curved layouts, such as layouts 4 and 5, likely because open areas temporarily collect water between blocks. The flow velocity is positively proportional to the downstream discharge rates and inversely proportional to the depth, except in the case of layout 7, which has a compact pattern and zigzag road network.

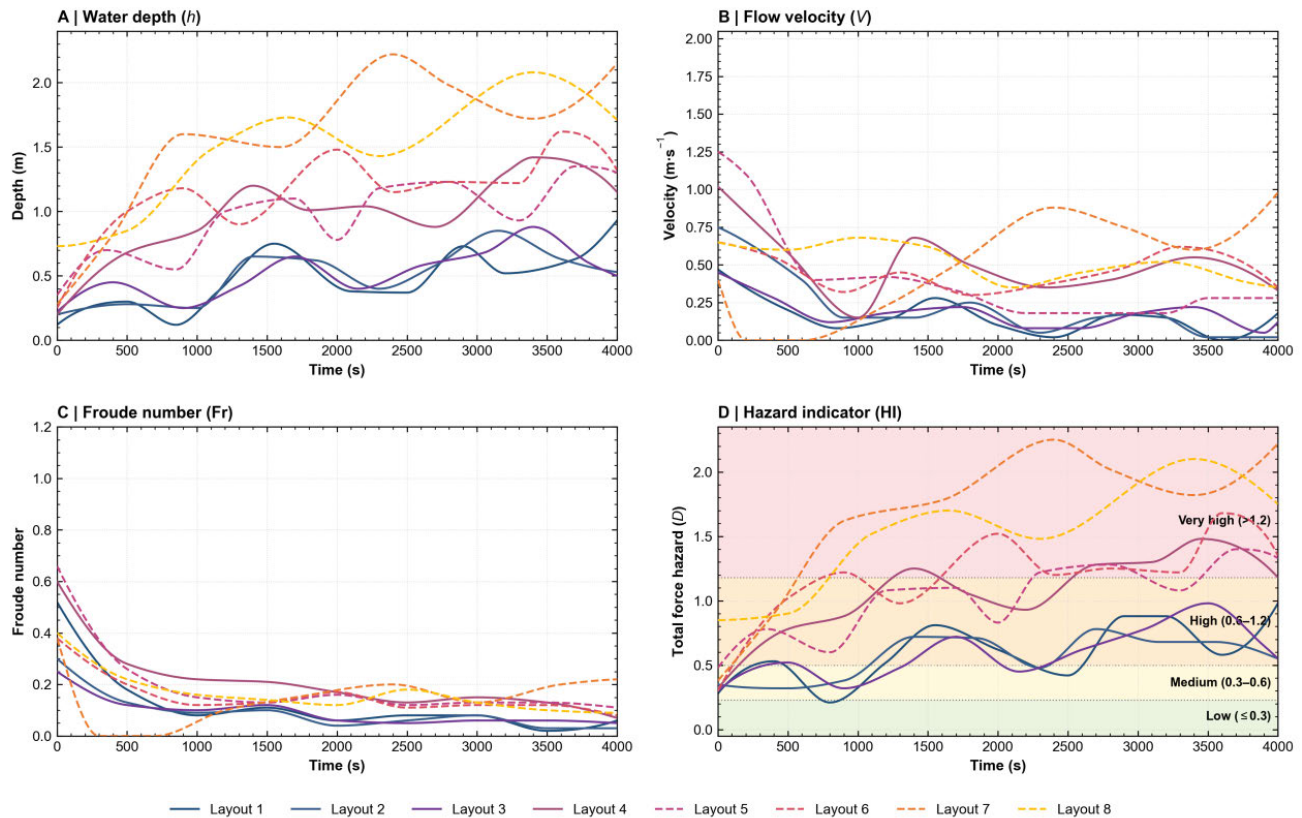


Fig. 8. Representative simulation outputs for the eight urban layouts, illustrating spatial differences in water depth, flow velocity, and Froude number; source: own study

Urban layouts are compared in Figure 9, focusing on critical flood-related metrics, across different time intervals. Layout 7 exhibits significant fluctuations in these parameters, with notable peaks such as a 2.10 m depth at the 800 time interval, indicating potential high-risk flood areas. In contrast, layouts 2 and 3 maintain generally lower depth and velocity values but show marked increases in hazard and discharge at specific times. Layout 6 demonstrates a unique stable depth and velocity measurement pattern coupled with widely varying hazard levels. At the same time, layout 8 presents moderate values across all metrics with occasional significant spikes, such as a hazard value of 1.12 at the 3500 time point. The diverse behaviours observed across eight layouts underscore the complexity of urban flood dynamics and emphasise the critical need for tailored, context-specific flood management strategies that address the unique characteristics of each urban configuration, thereby enhancing overall flood resilience. Overall, the results demonstrate clear variations in flood behaviour across different urban configurations, supporting the hypothesis that urban morphology significantly influences flood dynamics under identical hydraulic conditions.

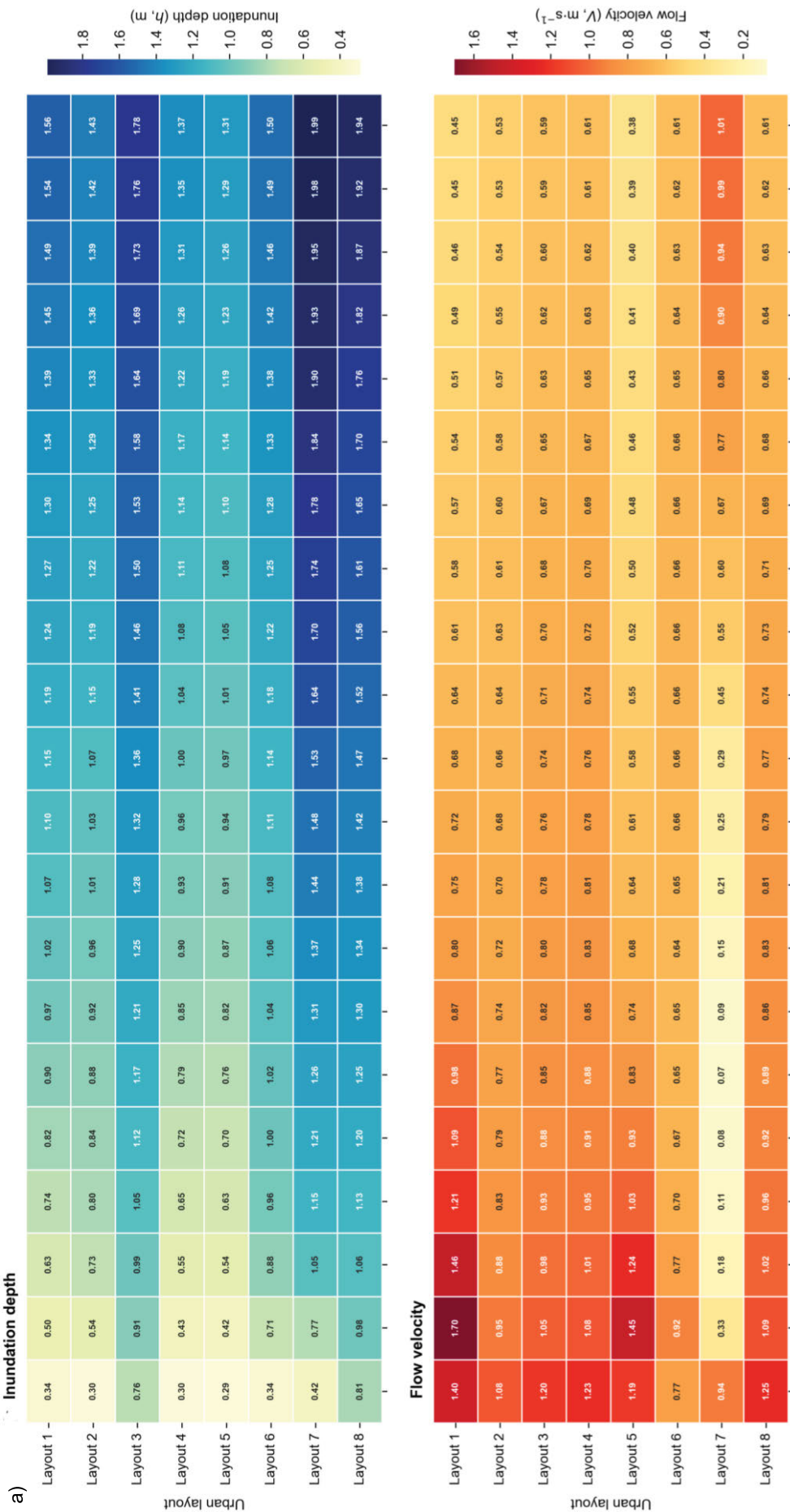
STATISTICAL ANALYSIS FOR URBAN FORM INDICATORS AND WATER DEPTHS

The relationship between urban characteristics and flood severity

The regression analysis illustrated in Figure 10 offers a detailed examination of the relationship between urban design parameters and flood characteristics. Regression coefficients (a_i) were calculated for an inundation indicator ($y = \Delta h_{90}$), derived from

the 90th percentile of computed water depths along the upstream boundaries of the study area. Positive regression coefficients indicate that higher values of the corresponding variable are associated with higher water depths and increased flood hazard. For instance, a positive coefficient associated with building density suggests that higher building concentrations are associated with deeper flooding within the simulated scenarios. Conversely, negative coefficients indicate that higher parameter values are associated with lower water depth. The predictive capabilities of this regression model are evidenced by its accuracy in estimating Δh_{90} values, exhibiting a mean absolute error of 3.1 cm and a root mean square error of 3.9 cm. These error metrics are notably low compared to the mean Δh_{90} value of 28.3 cm, representing less than 14% of this mean value, thereby underscoring the model's high precision and reliability.

Based on regression analysis, the urban characteristics can be ranked from most influential to least influential as follows: parcel setbacks (−0.64), mean parcel area (0.41), street orientation (−0.21), street width (−0.12), street length (−0.09), and street curvature (0.08). Parameter 1 (Pa. 1) shows a positive association with inundation depth, indicating that larger parcels are associated with higher flood depths. This association may be related to the higher proportion of impervious surfaces typically observed in larger parcels, which can limit infiltration and increase runoff under the imposed hydraulic conditions. Additionally, larger parcels create longer flow paths for water, leading to greater accumulation in certain areas. Conversely, parallel setbacks have a negative correlation with inundation depth. Increased setbacks are associated with greater open space between buildings, which may facilitate improved water disper-



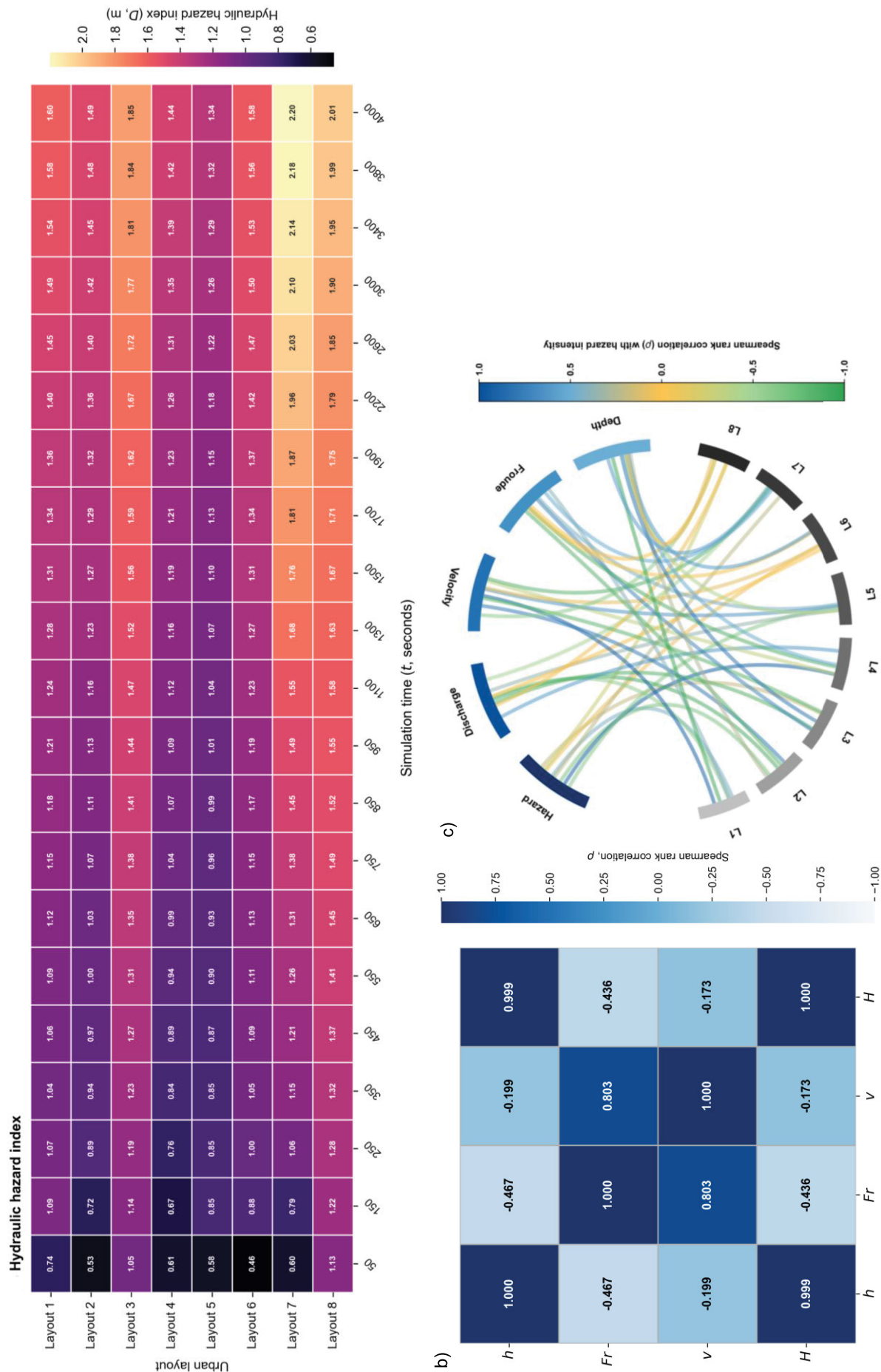


Fig. 9. Temporal evolution and correlation of flood flow characteristics across the eight urban layouts: a) heatmaps showing the evolution of inundation depth (h), flow velocity (v), and hydraulic hazard index (H) over the simulation period for each urban layout (layouts 1-8), b) Spearman rank correlation (ρ) matrix among water depth, Froude number (Fr), flow velocity, and hazard, c) chord diagram illustrating the correlations between flow variables across the eight urban layouts (L1-L8); source: own study

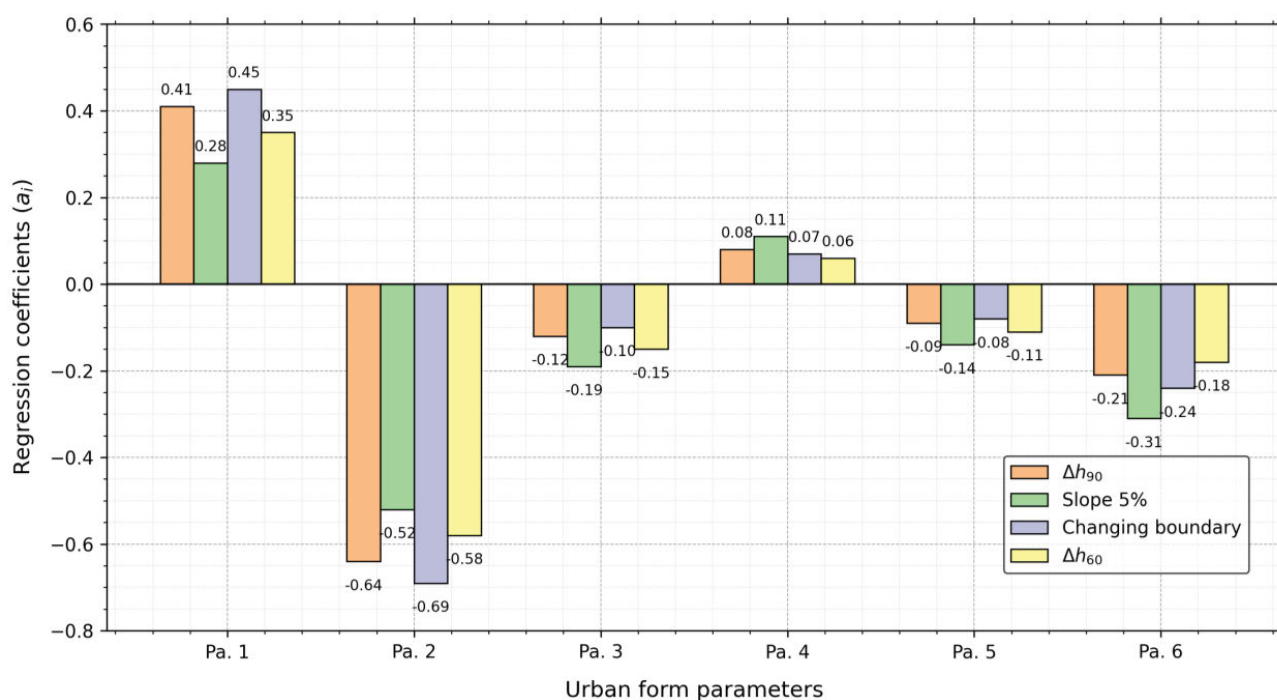


Fig. 10. Correlation coefficients illustrating the sensitivity of flood severity to urban form parameters; the analysis includes three alternative scenarios: i) changing the flood severity indicator from 90th percentile of computed water depths along the upstream boundary of the domain (Δh_{90}) to 60th percentile of computed water depths along the upstream boundary of the domain (Δh_{60}), ii) increasing the terrain slope to 5%, iii) modifying boundary conditions; Pa. = parameter; source: own study

sion and create natural drainage pathways, thus reducing flood depths. Street characteristics also play a role in determining inundation water depth. Street orientation has a moderate negative correlation, suggesting that specific orientations may facilitate better drainage and reduce flood depths. Street width shows a slight negative correlation, which may be explained by the tendency of wider streets to function as preferential flow paths, reducing localised flooding. The street length has a minimal negative impact on inundation depth due to longer streets providing more opportunities for water to disperse or drain. Interestingly, street curvature has a slight positive correlation with flood depth, which could be attributed to curved streets potentially creating areas where water accumulates or slowing down water flow in certain sections.

Urban flood dynamics under varying conditions: Insights from sensitivity analysis

A sensitivity analysis focuses on three key factors: changing flood severity from Δh_{90} to Δh_{60} (60th percentile of computed water depths along the upstream boundary of the domain), increasing slope to 5% in the same direction, and changing boundary conditions, as shown in Figure 11. The first scenario explored the impact of changing flood severity from Δh_{90} to Δh_{60} . This adjustment represents a shift from analysing extreme flood events to more moderate ones, typically resulting in lower inundation depths. Mean parcel areas and setbacks have slightly reduced impacts because they can manage extreme flooding less in moderate events. Street width becomes more influential as it plays a crucial role in conveying water during such events. The new regression values for this scenario are: Pa. 1 (0.35), Pa. 2 (-0.58), Pa. 3 (-0.15), Pa. 4 (0.06), Pa. 5 (-0.11), and Pa. 6 (-0.18). In the second scenario, increasing slope is associated with changes in flood dynamics by enhancing gravity-driven drainage, which

can reduce overall inundation depths (Fig. 12). The relative importance of urban characteristics also changes; accordingly, mean parcel area and parcel setbacks become less significant as the slope facilitates faster drainage, reducing their relative impact. Street width gains importance due to its effect on water velocity on inclined surfaces, while street curvature becomes more influential as it affects localised water accumulation or acceleration on slopes. The new regression values are: Pa. 1 (0.28), Pa. 2 (-0.52), Pa. 3 (-0.19), Pa. 4 (0.11), Pa. 5 (-0.14), and Pa. 6 (-0.31).

The third scenario involves changing boundary conditions by modifying inflow hydrographs, significantly altering water propagation through urban areas. This change was associated with a higher relative contribution of mean parcel area and parcel setbacks, as these factors become critical in managing altered inflow patterns. Street width became slightly less influential due to the dominance of boundary-driven flood dynamics. In contrast, street orientation gained importance as it interacted with new flow patterns created by boundary changes. The new regression values are: Pa. 1 (0.45), Pa. 2 (-0.69), Pa. 3 (-0.10), Pa. 4 (0.07), Pa. 5 (-0.08), and Pa. 6 (-0.24).

The sensitivity analysis confirms that variations in flood severity, terrain slope, and boundary conditions modify the relative influence of urban form parameters, while parcel-related characteristics consistently exhibit strong explanatory power across scenarios.

Predictive performance of urban characteristics in flood depth estimation

The impact of the six selected urban characteristics on computed water depths exhibited notable variation, prompting a comparison of the predictive capabilities of regression models based on either all urban characteristics (variables from Pa. 1 to Pa. 6) or solely the most influential ones. The effectiveness of each regression

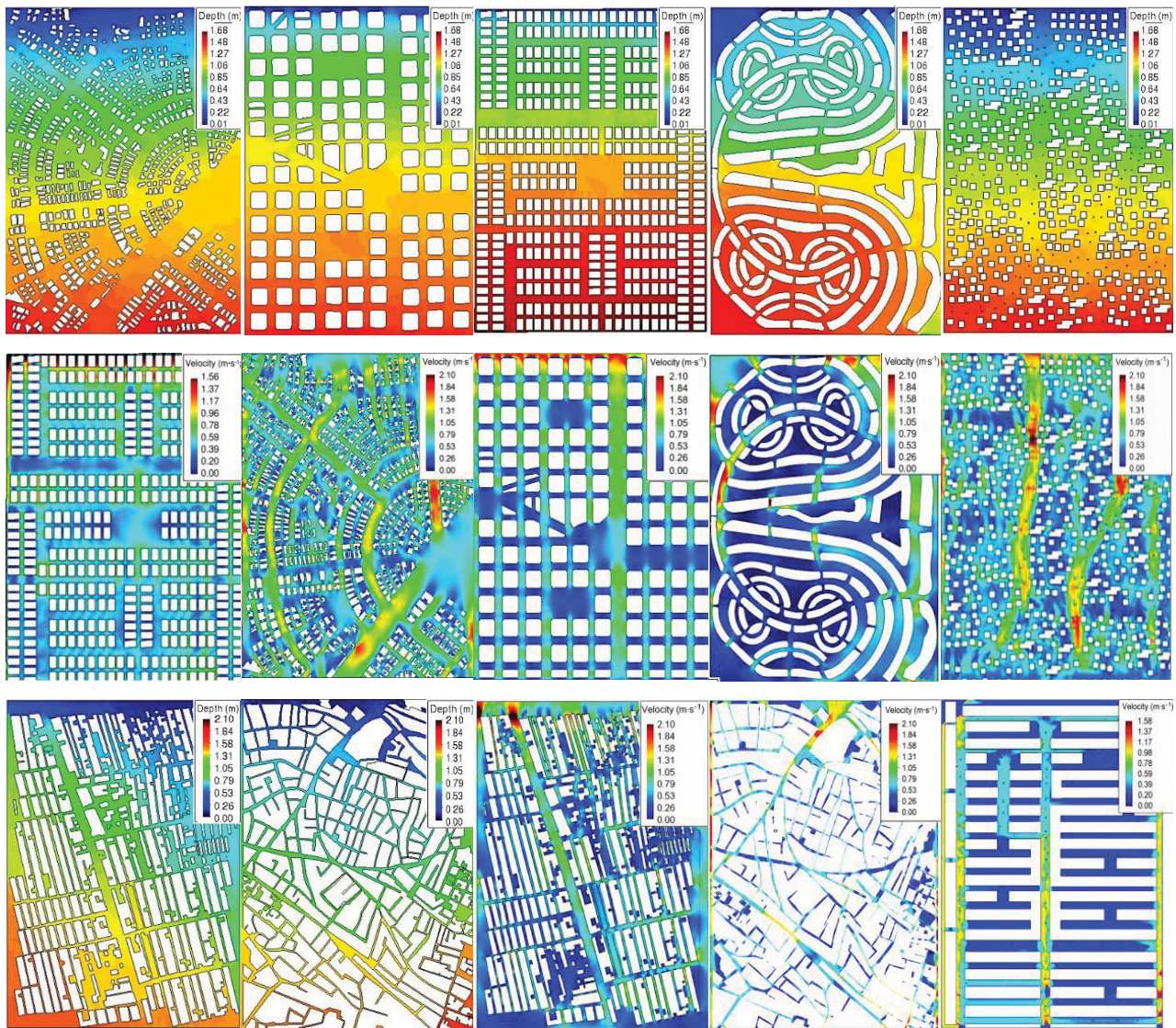


Fig. 11. Flood simulation results for the urban layouts under a terrain slope of 5%, illustrating changes in flood depth, velocity, and hydraulic behaviour compared with the baseline scenario; source: own study

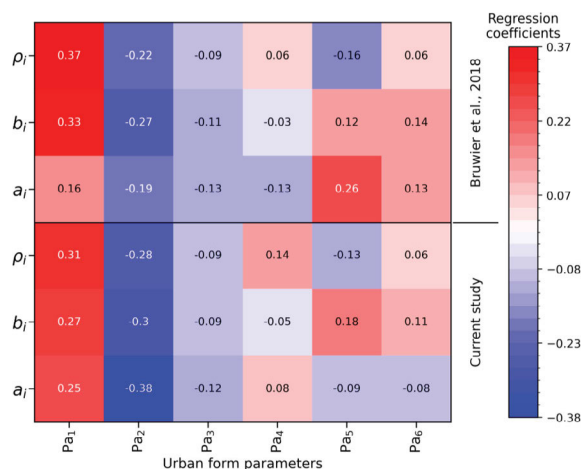


Fig. 12. Comparison of regression-based coefficients for urban form parameters; the upper panels show the results reported by Bruwier *et al.* (2018), the lower panels present the results of the current study; a_i = standardised regression coefficients, b_i = logarithmically transformed coefficients, ρ_i = Pearson correlation coefficients normalised to enable comparison of the relative influence of each parameter on Δh_{90} ; Pa. = parameters; source: own study

model was evaluated using the resulting root mean square error (*RMSE*) as a key metric. Utilising only parcel setbacks (Pa. 2) in linear regression resulted in an *RMSE* of 3.61 cm, approximately 65% higher than the *RMSE* obtained from multiple linear regression (MLR) that included all variables. Conversely, predicting Δh_{90} using the three most influential variables (Pa. 1, Pa. 2, and Pa. 6) yielded an accuracy of 2.34 cm, closely matching the accuracy achieved with all six variables, measured at 2.23 cm. Given the limited number of urban configurations ($n = 8$), the regression analysis is employed as an exploratory sensitivity assessment rather than a predictive model intended for statistical generalisation. It should be noted that the regression analysis is based on simulation outputs and is intended to assess statistical associations and relative sensitivities rather than to establish direct causal relationships.

To further investigate whether the modelling approach influenced these findings, alternative methods were compared with the results from Bruwier *et al.* (2018). This study selectively weighed six out of nine indicators from the previous research to align with our specific study parameters. Initially, a power law model was employed to predict Δh_{90} using all parameters from

Pa. 1 to Pa. 6, expressed as $\frac{y}{y_{\max}} = b_0 \prod_{i=1}^6 \left(\frac{pa_i}{pa_{i,\max}} \right)^{b_i}$ (where: b_0 – b_6 = coefficients that require calibration, y = the predicted response variable (Δh_{90}), $y_{\max}$ = the maximum (or reference) value of the response variable, used to normalise y , i = the index representing each parameter, running from 1 to 6 (corresponding to Pa. 1–Pa. 6), p = shorthand/notation indicating the parameter set used in the model, a_i = the value of the i -th urban form parameter (e.g., mean parcel area, building setback, street length, etc.), $a_{i,\max}$ = the maximum value of the i -th parameter, used to normalise a_i , b_i = the calibrated coefficient (exponent) for the i -th parameter, indicating its relative influence on the response variable). Although the coefficients b_i do not correspond directly to the parameters a_i , their relative magnitudes indicate which variables exert greater influence on Δh_{90} . Additionally, Pearson correlation coefficients (additionally, Pearson correlation coefficient – ρ_i) were calculated to assess the relationships between variables independently of any specific model selection. The estimation of coefficients b_i involved applying a logarithmic transformation to the variables x_i and y , yielding the equation $\ln\left(\frac{y}{y_{\max}}\right) = \ln(b_0) + \sum_{i=1}^6 b_i \ln\left(\frac{pa_i}{pa_{i,\max}}\right)$. In Figure 12, a comparative analysis of coefficients a_i , b_i and ρ_i with each set normalised so that the sum of their absolute values equals one. In this study, the coefficients a_i represent standardised regression coefficients obtained from the multiple linear regression model, indicating the relative contribution of each urban parameter to Δh_{90} . The coefficients b_i correspond to parameters derived from the logarithmically transformed power-law model, reflecting the influence of each variable under nonlinear scaling. The Pearson correlation coefficients ρ_i quantify the strength and direction of the linear association between each parameter and Δh_{90} independently of model structure.

The analysis conducted through three distinct methodologies consistently highlights the significant impact of variables Pa. 1 and Pa. 2, associated with building coverage, on the flood severity indicator Δh_{90} . These approaches uniformly demonstrate a strong correlation or substantial influence of these variables on flood dynamics. Notably, the coefficients attributed to Pa. 2 and Pa. 3 exhibit consistently negative values of considerable magnitude across all methodologies, reinforcing the robustness of the conclusion that these two variables play a crucial role in governing water flow within urban environments. This finding aligns with previous research, which posits that building coverage primarily indicates an urban area's storage capacity. Such agreement across multiple analytical approaches supports the robustness of these results within the tested configurations and highlights their relevance for urban flood management.

To further evaluate the stability of the regression estimates, a leave-one-out cross-validation (LOOCV) procedure was conducted. The mean LOOCV RMSE was 2.41 cm, compared to 2.23 cm for the full model. In addition, the relative ranking of influential parameters remained stable across iterations. These results indicate that the identified parameter importance is not driven by overfitting despite the limited sample size.

It should be noted that the simulations are not calibrated against a specific real-world flood event, as the primary objective of this study is to assess the relative influence of urban morphological characteristics under controlled hydraulic conditions. Therefore, the results should be interpreted as comparative rather than predictive. To ensure robustness, model outputs were evaluated using statistical error metrics and cross-validation

techniques, and were also compared with findings from previous studies. This approach is consistent with established parametric and sensitivity-based studies in urban flood modelling, where the objective is to isolate governing processes rather than reproduce site-specific events.

DISCUSSION

CONSISTENCY OR MISMATCH WITH PREVIOUS STUDIES ON MICRO-SCALE URBAN FORM AND FLOW CHARACTERISTICS

The statistical analysis reveals that mean parcel size, setbacks, and street orientation are the most significant determinants of flow characteristics. These findings align with the results reported by Tomiczek et al. (2016), Bruwier et al. (2018), Bruwier et al. (2020), and Dong et al. (2021). Road networks aligned with downstream orientation play a crucial role in discharge management, consistent with Mohammed (2021). Furthermore, the effectiveness of a network of minor streets oriented downstream in flood management – compared to a few significant thoroughfares – can be attributed to several factors. This configuration allows for better distribution of floodwater across the urban landscape, preventing the concentration of large volumes of water in specific areas. Additionally, this network approach reduces water flow velocity, which can minimise erosion and infrastructure damage. It also mimics natural drainage patterns, facilitates more efficient water evacuation from urban areas, and provides alternative flow paths if specific routes become obstructed. Implementing building setbacks and open spaces improves storage capacity and enhances flow dispersion, which reduces local water accumulation and flood depth. This strategy substantially reduces the number of buildings exposed to hazards while effectively mitigating floodwater volume and velocity. Moreover, setbacks facilitate the incorporation of green infrastructure elements, enhancing the aesthetic value of the urban landscape and providing additional flood mitigation benefits. This study extends previous research by providing a comparative and integrated assessment of multiple urban form parameters under controlled hydraulic conditions. In particular, it identifies the relative importance of parcel size, building setbacks, and street network configuration, offering a clearer interpretation of their combined influence on flood dynamics.

INSIGHTS FOR WATER-SENSITIVE URBAN FORM

According to previous flow scenarios, layouts 2 and 3 are the most effective in controlling the flow processes at light slopes. These layouts are characterised by semi-straight networks oriented downstream and aligned either west-east or north-south, separated by uniformly sized building blocks. Building setbacks act as narrow internal corridors that support downstream flow and reduce pressure on the main road network. This design contrasts with looped or closed-end road networks, which prolong the presence of water and complicate safe disposal. Four conceptual models that are similar in network planning but differ in permeable surface, space distribution, and configuration are illustrated in Figure 13. Options 1 and 4 focus solely on street porosity through blue-green networks or permeable material-based infiltration, effectively integrating water elements into street island landscaping and merging the aesthetic and functional aspects of urban road design (Ahmed,

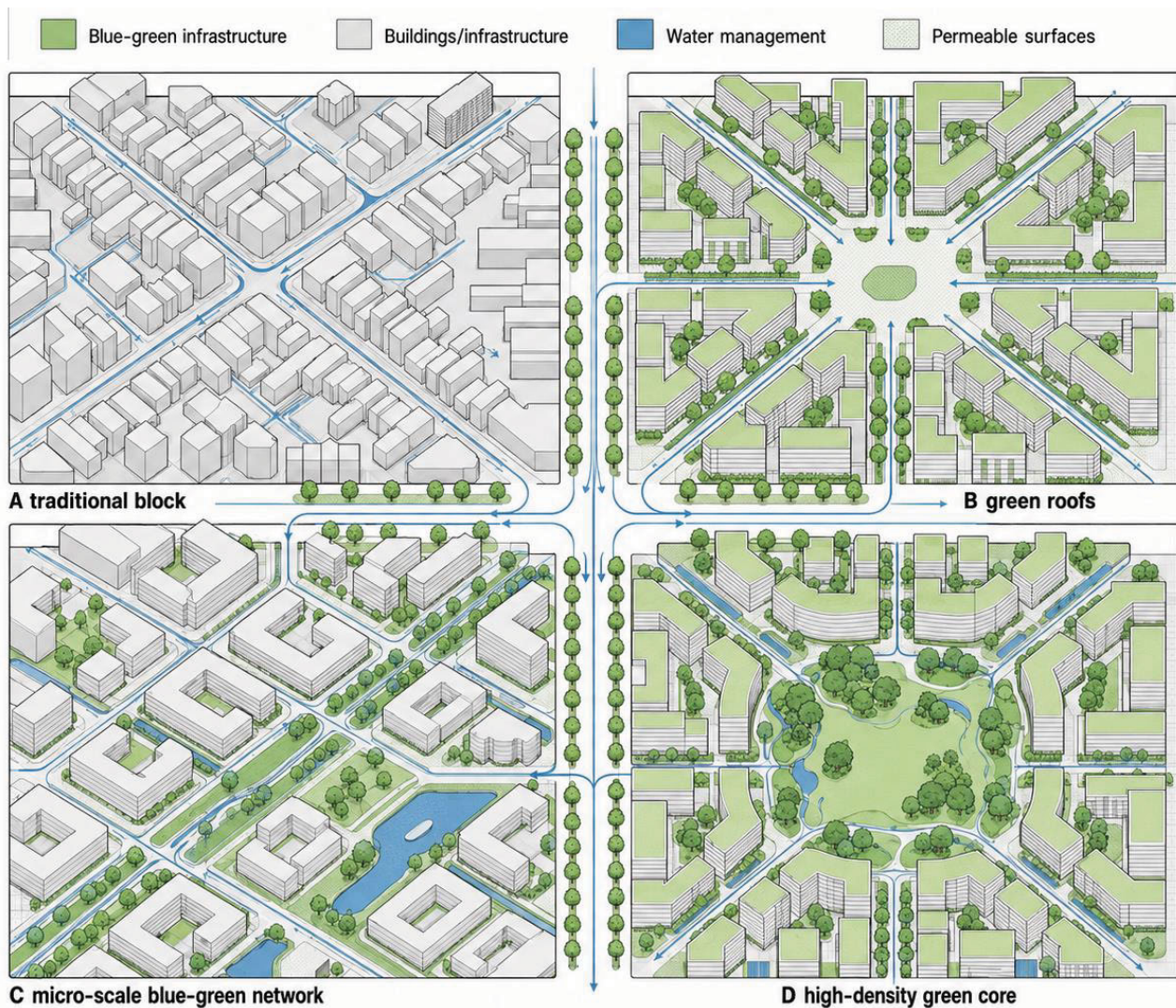


Fig. 13. Conceptual design guidance for micro-scale water-sensitive urban forms, illustrating planning strategies that enhance urban flood resilience through street configuration, parcel layout, and open space integration; source: own study

Meenar and Alam, 2019). Options 2 and 3, the central and pocket park scenarios, integrate green spaces as multi-directional rainwater collection basins. Beyond their primary role in flood mitigation, these spaces enhance urban resilience (Panno *et al.*, 2017). Figure 14 illustrates a grid-based urban layout integrated with a green network, demonstrating how structured street geometry and open spaces can support water-sensitive urban design and urban flood resilience.

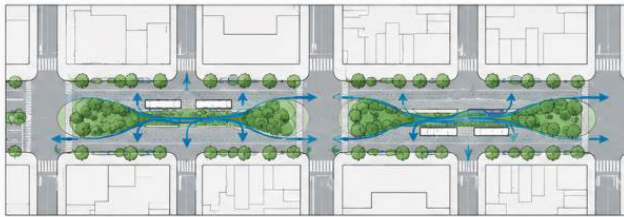
URBAN PLANNING AND ADMINISTRATIVE GUIDELINES FOR FLOOD-RESILIENT URBAN FORM

This study examines the micro-scale interconnections between urban form characteristics and flow dynamics through interdisciplinary approaches. It emphasises the significance of designing interconnected street networks, considering building setbacks, and prioritising the preservation of green spaces to mitigate flood risks and enhance overall resilience. These outcomes can provide co-benefits that extend beyond flood reduction by addressing socio-economic challenges, improving ecosystem services and air quality, enhancing urban livability, creating attractive recreational spaces for residents and visitors, and fostering a transition from self-

protection to harmony with nature. These recommendations can be effectively implemented by integrating administrative and urban planning mechanisms. First, administrative frameworks can establish a legal foundation by updating zoning laws and building codes to prioritise the development of green spaces. Furthermore, cities can create dedicated offices or departments responsible for overseeing the implementation of nature-based solution practices. Community engagement in planning, public-private partnerships, and effective coordination with stakeholders can facilitate urban form modifications that are socially acceptable and culturally appropriate. This collaborative approach ensures that local needs and priorities are addressed, fostering a sense of ownership and responsibility among community members regarding flood resilience initiatives.

On the other hand, urban planning mechanisms can strengthen sustainable urban intensification and expansion by enacting long-term building control legislation that regulates setbacks, land use, height, and building coverage ratios. Additionally, city planners should adopt blue-green infrastructure strategies to reconfigure flood-prone communities and redevelop brownfields – such as fragile lands, buffer zones, and urban farming areas – as protective measures against flooding. Finally,

1. Infiltration streets



Using highly porous materials and street-island landscaping to integrate water directly into transit corridors.

3. Central parks



A large, unified green space acting as a multi-directional primary rainwater collection basin.

2. Blue-green networks



Connecting street porosity with continuous ecological corridors between parcels.

4. Pocket parks



Decentralised micro-basins embedded within the parcel grid for localised rapid storage.

Fig. 14. Example of a grid-based urban layout integrated with a green network, illustrating how structured street geometry and open spaces can support water-sensitive urban design; source: own study

adopting a holistic strategy that combines structural and non-structural measures is crucial, as fragmented implementations may yield unfavourable outcomes in response to unpredictable climate fluctuations. These recommendations require leveraging the expertise of urban planners, urban ecologists, horticulturalists, and landscape planners to ensure a comprehensive perspective that transcends land ownership boundaries.

The integration of urban planning and administrative mechanisms derived from the study and highlights of their combined role in enhancing flood resilience are summarised in Figure 15.

Despite these practical implications, the study remains subject to methodological limitations that should be acknowl-

edged. This study does not include validation against observed flood events or physical experiments. The objective is to isolate the relative influence of urban morphological characteristics under controlled hydraulic conditions rather than to reproduce site-specific scenarios. Therefore, the results should be interpreted as comparative and process-oriented. While this limits direct predictive applicability, it allows clearer identification of the role of urban form independent of external variables. In addition to the parameters analysed in this study, other factors such as surface permeability, green-blue infrastructure, rainfall variability, and topographic heterogeneity may also influence flood dynamics and should be considered in future research.

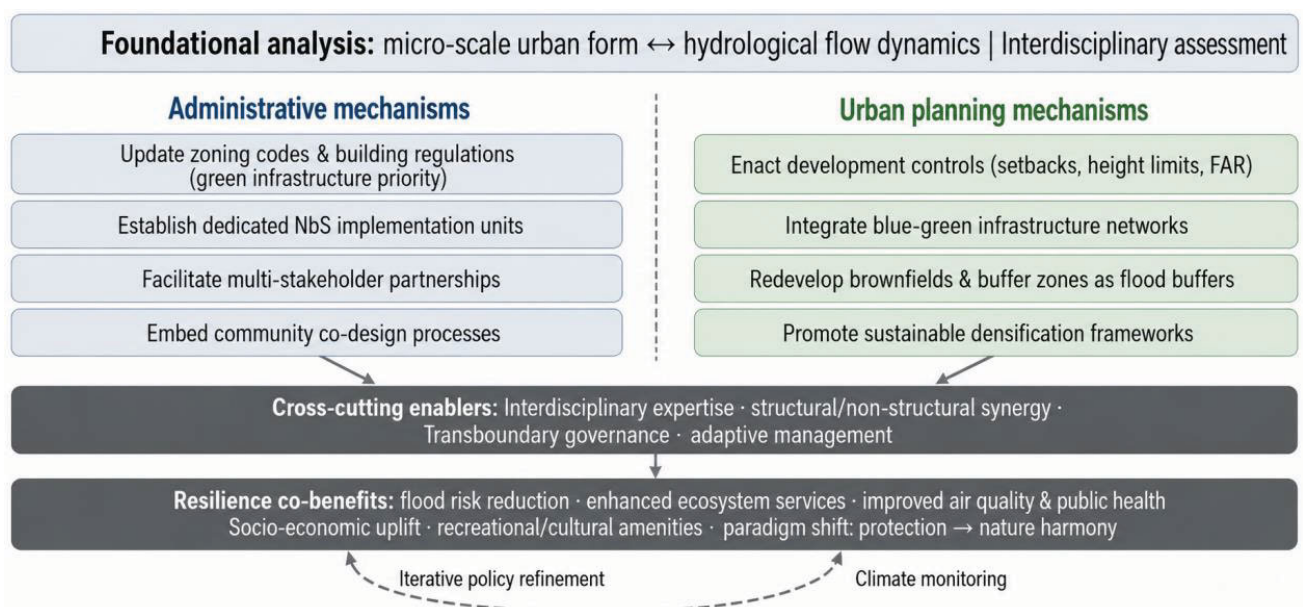


Fig. 15. Integrated framework linking urban planning and administrative mechanisms to flood resilience co-benefits based on the study findings; FAR = floor area ratio; NbS = nature-based solutions; source: own study

CONCLUSIONS

This study assessed the influence of diverse and complex urban morphologies on flow characteristics. A set of eight varied urban forms comprising four procedurally generated and four realistic configurations was generated over a 2.0 km² area. Two-dimensional hydraulic simulations were conducted using obstacle representation through mesh voids and uniform roughness conditions for each urban form under identical hydraulic conditions to evaluate the influence of six urban parameters: street orientation, street curvature, street length, street width, mean parcel area, and building setbacks on flood characteristics, including flow velocity, water depth, flood hazard, and discharge. A slight slope of 1% was selected to isolate the effect of urban planning parameters while minimising the influence of other factors. The findings can be summarised in four main points.

- The simulation results indicated that out of six parameters, the mean parcel area, building setbacks (which directly control building coverage), and street orientation are the most significant urban form factors influencing flow dynamics at a slight slope. These results match previous findings, which showed that building coverage, mean parcel area, and building side setbacks are the most effective in the 90th percentile of computed water depths along the upstream boundary of the domain (Δh_{90}).
- For the tested configurations, increasing building setbacks was associated with reduced water depth and, accordingly, hazard levels both upstream and downstream compared to layouts without setbacks.
- Grid street networks aligned with the prevailing slope were associated with lower water depths and improved outflow discharge. Conversely, curved streets and zigzag roads were associated with localised bottlenecks and higher water accumulation and extended disposal times, thereby increasing water depth and associated hazards in those areas.
- In sensitivity analyses, an increased slope of 5% confirmed the significant impact of street curvature and orientation on water depth compared to the slight slope of 1%. These findings underscore the importance of block-scale building arrangements and conveyance porosity in developing effective flood management strategies.

The analysis of these scenarios demonstrates that urban flood dynamics are sensitive to changes in flood severity, terrain slope, and boundary conditions. Each scenario modifies the relative importance of urban characteristics, while parcel-related features (mean parcel area and mean building setbacks, which impact building coverage) consistently maintain high significance across all conditions. These findings highlight the complexity of urban flood modelling and the need for adaptive urban planning and management strategies that account for varying flood scenarios and urban configurations.

These interdisciplinary findings provide valuable insights into the complex interactions between block-scale urban design and flood resilience. Further research is needed to gain a more comprehensive understanding of obstacles such as vehicles, storage and conveyance porosity, and fluctuating hydraulic conditions like unsteady flood waves that impact flood dynamics. While grid street patterns may be more effective at quickly eliminating runoff than curved streets, this advantage diminishes in the absence of nearby downstream water bodies. Additional

studies should examine runoff patterns in various urban configurations, particularly in large agglomerated urban areas. This study primarily focuses on the characteristics of two-dimensional inundation flow; however, exploring three-dimensional aspects is essential for a more comprehensive understanding. Ultimately, future research can investigate the implications of integrating flood-resilient urban planning models with other sectors, such as transportation and housing.

SUPPLEMENTARY MATERIAL

Supplementary material to this article can be found online at: https://www.jwld.pl/files/Supplementary_material_70_Mabrouk.pdf.

CONFLICT OF INTERESTS

All authors declare that they have no conflict of interests.

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